

ELARIS
AI-Powered Academic Assistant

Project Documentationa

Final Year Engineering Project
Department of Computer Science / Information Technology

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1. Project Introduction

What is ELARIS?

ELARIS is an AI-powered academic assistant. It helps students organize study material, prepare for exams, and get instant academic support using Artificial Intelligence.

Purpose of the Project

The purpose of ELARIS is to give students a single platform where they can store notes, ask questions, and get AI-generated help for their studies.

Vision

To become the most trusted AI study companion for students, making learning simple, organized, and accessible to everyone.

Mission

To use Artificial Intelligence to reduce the time students spend searching for study material and help them focus more on actual learning.

Target Users

- College and university students
- Students preparing for exams
- Students who struggle to organize notes
- Students who need quick academic help

Main Objectives

- Provide a centralized place for study material
- Use AI to answer academic questions
- Help students prepare for exams faster
- Reduce time spent searching for notes and resources
- Make learning simple and organized

2. Problem Statement

Students face many challenges while studying. The table below explains the main problems and how ELARIS solves them.

Problem	Explanation	How ELARIS Solves It
Scattered Study Material	Students keep notes on WhatsApp, Drive, and different apps.	ELARIS stores all notes in one place.
Unreliable Notes	It is hard to know which notes are correct or complete.	ELARIS lets students access verified, organized notes.
No Central PYQ Bank	Previous Year Questions are hard to find in one place.	ELARIS provides a dedicated PYQ Bank.
Time Wasted Searching	Students spend hours looking for the right resource.	ELARIS uses smart search to find material instantly.
Difficult Topics	Some topics are hard to understand from notes alone.	AI Assistant explains topics in simple terms.
Limited Support Outside Class	Teachers are not available all the time for doubts.	ELARIS gives 24x7 AI-based academic support.
Confusing Assignment Planning	Students struggle to plan and complete assignments on time.	Assignment Assistant helps plan and track tasks.

3. Proposed Solution

How ELARIS Works

- Students upload notes, PDFs, and study material to the platform.
- The system reads and processes the uploaded content using AI.
- Students can ask questions and get answers based on their own notes.

- ELARIS also generates summaries, flashcards, and quizzes automatically.

What Makes It Different

- Answers come from the student's own study material, not random internet content.
- All academic tools are available on one platform.
- Uses AI models that run locally, which keeps the system fast and cost-effective.

Why It Is Useful

- Saves study time
- Improves exam preparation
- Makes learning more organized and personal

4. Technology Stack

The table below lists the technologies used in ELARIS and the reason each one was selected.

Technology	Purpose	Why Selected
React.js	Builds the user interface	Fast, flexible, and widely used for modern web apps
Tailwind CSS	Styles the application	Speeds up UI design with ready-to-use classes
FastAPI	Powers the backend server	Lightweight, fast, and works well with Python and AI
Python	Main backend programming language	Best language for AI and machine learning tasks
PostgreSQL	Stores application data	Reliable and handles structured data efficiently

Technology	Purpose	Why Selected
JWT Authentication	Manages secure user login	Industry-standard method for secure sessions
Ollama	Runs AI models locally	Allows AI to run without heavy cloud cost
Llama 3.2 / Qwen 2.5	AI models used for academic assistance	Open-source models suitable for local deployment
ChromaDB	Stores AI-searchable note data	Vector database designed for AI-based search
PyMuPDF	Extracts text from PDF notes	Reliable library for reading PDF content
Git	Tracks code changes	Standard tool for version control
GitHub	Hosts and manages project code	Enables collaboration and code backup
Docker	Packages the application	Makes the app easy to deploy anywhere

5. System Architecture

ELARIS is built using a few core components that work together. Each component and its role is explained below.

Frontend

- **What it does:** The part of the app students see and interact with.
- **Why it is needed:** Needed to give students an easy-to-use interface.

Backend

- **What it does:** Handles all the logic and processing behind the scenes.
- **Why it is needed:** Connects the frontend, database, and AI model together.

Database

- **What it does:** Stores student data, notes, and app information.
- **Why it is needed:** Needed to save and retrieve information reliably.

Authentication

- **What it does:** Manages secure student login and access.
- **Why it is needed:** Protects student data and personal accounts.

AI Model

- **What it does:** Understands notes and answers student questions.
- **Why it is needed:** Core part of ELARIS that provides academic help.

File Processing

- **What it does:** Reads and extracts text from uploaded notes and PDFs.
- **Why it is needed:** Needed so AI can understand uploaded material.

Vector Database

- **What it does:** Stores notes in a format AI can search quickly.
- **Why it is needed:** Allows fast and accurate AI-based search.

Search

- **What it does:** Helps students find notes, PYQs, and material fast.
- **Why it is needed:** Saves time when looking for specific content.

Admin Panel

- **What it does:** Lets admins manage users and content.
- **Why it is needed:** Needed to maintain and monitor the platform.

6. MVP Features

ELARIS includes the following core features in its MVP (Minimum Viable Product) version.

Authentication

- **Purpose:** Allow students to securely create an account and log in.

- **What it does:** Provides sign-up, login, and secure session management.

Benefits:

- Keeps student data safe
- Gives each student a personal space
- Prevents unauthorized access
- **Technologies used:** FastAPI, JWT Authentication, PostgreSQL

Student Dashboard

- **Purpose:** Give students a central view of their academic activity.
- **What it does:** Shows uploaded notes, recent activity, and quick access to tools.

Benefits:

- Saves time navigating the app
- Keeps everything organized
- Gives a quick overview of progress
- **Technologies used:** React.js, Tailwind CSS, FastAPI

AI Academic Assistant

- **Purpose:** Let students ask academic questions and get instant answers.
- **What it does:** Uses AI to answer questions based on uploaded notes or general topics.

Benefits:

- Provides 24x7 academic help
- Explains difficult topics simply
- Reduces dependency on external tutors
- **Technologies used:** Ollama, Llama 3.2 / Qwen 2.5, ChromaDB

Notes Hub

- **Purpose:** Give students one place to upload and access all notes.
- **What it does:** Allows uploading, organizing, and viewing notes and PDFs.

Benefits:

- Ends scattered storage of notes
- Easy access anytime
- Keeps notes organized by subject
- **Technologies used:** React.js, PyMuPDF, PostgreSQL

AI Note Summarization

- **Purpose:** Help students quickly understand long notes.
- **What it does:** Uses AI to generate short summaries of uploaded notes.

Benefits:

- Saves reading time
- Helps with quick revision
- Improves understanding of long topics
- **Technologies used:** Ollama, Llama 3.2 / Qwen 2.5

Flashcard Generator

- **Purpose:** Help students revise key concepts quickly.
- **What it does:** Automatically creates flashcards from uploaded notes.

Benefits:

- Makes revision faster
- Improves memory retention
- Saves manual effort of making flashcards
- **Technologies used:** Ollama, Llama 3.2 / Qwen 2.5, ChromaDB

Quiz Generator

- **Purpose:** Help students test their knowledge.
- **What it does:** Generates quiz questions automatically from notes or topics.

Benefits:

- Improves exam readiness
- Makes self-testing easy
- Highlights weak areas

- **Technologies used:** Ollama, Llama 3.2 / Qwen 2.5

PYQ Bank

- **Purpose:** Give students access to previous year questions.
- **What it does:** Stores and organizes PYQs by subject and year.

Benefits:

- Centralized PYQ access
- Better exam preparation
- Saves time searching online
- **Technologies used:** PostgreSQL, React.js

Assignment Assistant

- **Purpose:** Help students plan and complete assignments on time.
- **What it does:** Assists with organizing tasks and tracking assignment deadlines.

Benefits:

- Reduces last-minute stress
- Improves time management
- Keeps assignments organized
- **Technologies used:** FastAPI, PostgreSQL, React.js

Global Search

- **Purpose:** Let students find any content quickly across the platform.
- **What it does:** Searches notes, PYQs, and flashcards from one search bar.

Benefits:

- Saves time
- Improves usability
- Gives instant access to relevant content
- **Technologies used:** ChromaDB, FastAPI

Admin Panel

- **Purpose:** Allow admins to manage the platform.
- **What it does:** Provides tools to manage users, content, and system health.

Benefits:

- Keeps platform organized
- Helps monitor usage
- Enables quick issue resolution
- **Technologies used:** React.js, FastAPI, PostgreSQL

Notifications

- **Purpose:** Keep students updated on important activity.
- **What it does:** Sends alerts for deadlines, updates, and reminders.

Benefits:

- Prevents missed deadlines
- Keeps students informed
- Improves engagement with the app
- **Technologies used:** FastAPI, PostgreSQL

7. Working Flow

The steps below explain how ELARIS works from login to learning.

1. Student logs in to ELARIS.
2. Student opens the dashboard.
3. Student uploads notes or a PDF.
4. The system extracts text from the uploaded file.
5. AI processes and understands the notes.
6. Processed notes are stored in the vector database.
7. Student asks a question in the AI Assistant.
8. AI searches the stored notes for relevant information.
9. AI generates a clear and simple answer.
10. Student continues learning with summaries, flashcards, or quizzes.

8. Advantages

- Saves study time
- Improves exam preparation
- Provides AI-powered learning support
- Keeps all notes organized in one place
- Enables faster searching of study material
- Gives personalized academic assistance
- Available 24x7 for student support
- Reduces stress around assignment deadlines
- Centralizes previous year questions
- Makes difficult topics easier to understand
- Encourages consistent revision through flashcards and quizzes
- Reduces dependency on scattered platforms like WhatsApp or Drive
- Easy to use, even for first-year students

9. Future Scope

ELARIS can be expanded further with the following features in future versions.

- Attendance management
- Timetable generation
- Resume builder
- Placement preparation tools
- Voice assistant support
- OCR support for handwritten notes
- Mobile application
- Multi-language support
- AI-based personalized recommendation system

10. Conclusion

- ELARIS solves real problems faced by students during their academic journey.
- It brings notes, PYQs, and AI support together on one platform.
- AI-powered tools like summarization, flashcards, and quizzes make learning faster.
- It saves students time by reducing the need to search across multiple platforms.
- The platform is simple to use, even for first-year students.
- ELARIS supports better exam preparation and stronger academic performance.
- It has strong potential for future growth with more student-focused features.